Suchita Sharma

suchitasharma2599@gmail.com | linkedin.com/in/suchitasharma99 | (857) 335-7669

SUMMARY

Data & AI Engineer with 3+ years building ELT pipelines, data models, and analytics-ready datasets with Python, SQL, dbt, Snowflake, and AWS. Turned messy, multi-source data (scanned PDFs, 100+ districts, 50K+ records) into star-schema tables, automated KPI monitoring, and 20% faster reporting, cutting manual work 60–70%. MS in Data Science, Northeastern University (2026).

PROFESSIONAL EXPERIENCE

AlchemyK12 **Boston, MA**
Data Engineer July 2026 – Present

- Build Python scrapers and pandas pipelines that standardize district strategic plans, state report cards, and scanned PDFs into a unified schema, covering 100+ districts across 2 states (3rd in progress) and replacing manual reformatting.
- Store cleaned datasets in S3 as partitioned Parquet and query them through AWS Glue and Athena, improving data processing speed by 30%.
- Manage pipeline infrastructure as code with Terraform and automated deployments via GitHub Actions, ensuring consistent releases across environments.
- Authored the team's data-source playbook, now the standard process for onboarding each new state, including the 3rd state currently in progress.

Machine Learning Intern September 2025 – December 2025

- Replaced manual validation across 10+ multi-state datasets with automated Python ingestion pipelines and schema-level quality gates, cutting manual validation time by 70% and making data uploads self-service.
- Diagnosed recurring failures in a production orchestration workflow caused by API misconfiguration and environment drift, then redesigned it as Airflow DAGs with validation checks and alerting, reducing pipeline failures by 70%.
- Resolved conflicting records across multi-source PostgreSQL schemas using deduplication, matching rules, and database constraints, improving data integrity across 50,000+ records.

Echo Labs **Boston, MA**
Data Engineer Intern May 2025 – August 2025

- Built 3 ELT pipelines across 5+ data sources using Snowflake, Airbyte, and dbt to get raw data into star-schema-aligned analytics tables, cutting manual intervention by 60%.
- Stood up monitoring across 5+ mission-critical KPIs with PySpark-based anomaly detection, cutting incident resolution time by 40%.
- Migrated the team from Firebase to Supabase (PostgreSQL) with Row-Level Security, rewriting 10+ slow queries.
- Restructured Snowflake warehouse layers into dbt dimensional models, speeding up reporting by 20%.

Flentas Technologies Pvt Ltd **Pune, India**
Cloud Engineer January 2021 – July 2023

- Owned infrastructure across 8 AWS services (EC2, S3, Lambda, RDS, VPC, IAM, ECS, EKS) for 15+ production applications, backed by proactive monitoring and on-call incident response.
- Introduced Docker-based CI/CD and Kubernetes orchestration, cutting release cycles by 50%, and coordinated release planning across 3 engineering teams to reduce conflicts and rollbacks by 40%.
- Audited spend across 3 AWS accounts, identified ~25% waste, and rightsized EC2 instances and S3 lifecycle policies with zero service disruption.

PROJECTS

Personalized Financial Recommendation System | Python, scikit-learn, dbt, PySpark, REST API
January 2026 – April 2026

- Designed a REST API layer and fact-dimension schema serving household financial recommendations, integrating three real-world datasets covering 13,886 households into a unified 74-feature backend..
- Ran and compared four clustering algorithms (K-Means, GMM, HDBSCAN, Agglomerative) on a 16-feature subset to find meaningful household archetypes for cold-start recommendation coverage.

Vani: Multi-Agent Research Assistant | Python, LangGraph, Vertex AI, FAISS, RAG, React
December 2025

- Built a Researcher → Reviewer → Synthesizer agent workflow over 100+ research documents, with evaluation checks that catch hallucinated or unsupported claims before they reach users.
- Optimized the retrieval path and session state management in a React + Vite interface to keep retrieval latency under 200ms under concurrent document loads.

EDUCATION

Northeastern University **Boston, MA**
Khoury College of Computer Sciences September 2023 – May 2026
Master of Science in Data Science

Savitribai Phule Pune University **Pune, India**
Smt Kashibai Navale College of Engineering August 2017 – May 2021
Bachelor of Engineering in Information Technology

SKILLS

Languages: Python, SQL, Shell/Bash, Java
Databases & Warehouses: Snowflake, PostgreSQL, Supabase, MySQL, Redis, Databricks
AI & LLM Tooling: LangChain, LangGraph, RAG, FAISS, OpenAI API, Vertex AI, scikit-learn
Data Engineering: dbt, pandas, Langfuse, Airflow, Airbyte, Apache Spark (PySpark), ETL/ELT, Data Modeling, Data Quality & Validation
Cloud & DevOps: AWS (S3, Glue, Athena, Lambda, EC2, RDS, ECS, EKS, IAM, VPC), Terraform, Docker, GitHub Actions, CI/CD